

Occupational Safety and Health (Classification, Labelling and Safety Data Sheet of Hazardous Chemicals) Regulations 2013

**SECTION 1: IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING**

**Product Identifier**

Perihalan Produk: **NATRIUM FOSFAT**  
Product Description: **Sodium phosphate, monobasic dihydrate**  
Cat No. : 271750000; 271750010; 271750025  
Synonyms Sodium dihydrogen phosphate dihydrate  
CAS No 13472-35-0  
Molecular Formula  $H_2 Na O_4 P \cdot 2 H_2 O$

**Relevant identified uses of the substance or mixture and uses advised against**

Recommended Use Laboratory chemicals.  
Uses advised against No Information available

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**SECTION 2: HAZARDS IDENTIFICATION**

**Classification of the substance or mixture**

**Label Elements**

**Hazard Statements**

**Storage**  
P403 - Store in a well-ventilated place

**Disposal**  
P501 - Dispose of contents/ container to an approved waste disposal plant

**Other Hazards**

This product does not contain any known or suspected endocrine disruptors

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## SECTION 3: COMPOSITION/INFORMATION ON INGREDIENTS

| Component                             | CAS No     | Weight % |
|---------------------------------------|------------|----------|
| Sodium phosphate, monobasic dihydrate | 13472-35-0 | >95      |
| Sodium phosphate, monobasic           | 7558-80-7  | -        |

## SECTION 4: FIRST AID MEASURES

### Description of first aid measures

#### Eye Contact

Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Get medical attention.

#### Skin Contact

Wash off immediately with plenty of water for at least 15 minutes. Get medical attention immediately if symptoms occur.

#### Ingestion

Clean mouth with water and drink afterwards plenty of water. Get medical attention if symptoms occur.

#### Inhalation

Remove to fresh air. Get medical attention immediately if symptoms occur.

#### Self-Protection of the First Aider

No special precautions required.

### Most important symptoms and effects, both acute and delayed

May be harmful if absorbed through skin. May cause adverse kidney effects. Ingestion may cause gastrointestinal irritation, nausea, vomiting and diarrhea.

### Indication of any immediate medical attention and special treatment needed

#### Notes to Physician

Treat symptomatically.

## SECTION 5: FIREFIGHTING MEASURES

### Extinguishing media

#### Suitable Extinguishing Media

Use extinguishing measures that are appropriate to local circumstances and the surrounding environment.

#### Extinguishing media which must not be used for safety reasons

No information available.

### Special hazards arising from the substance or mixture

Non-combustible. Keep product and empty container away from heat and sources of ignition. Thermal decomposition can lead to release of irritating gases and vapors.

#### Hazardous Combustion Products

Oxides of phosphorus.

### Advice for fire-fighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

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## SECTION 6: ACCIDENTAL RELEASE MEASURES

### Personal Precautions, Protective Equipment and Emergency Procedures

Use personal protective equipment as required. Ensure adequate ventilation. Avoid dust formation.

### Environmental precautions

Should not be released into the environment.

### Methods and Material for Containment and Cleaning Up

Sweep up and shovel into suitable containers for disposal. Avoid dust formation.

### Reference to Other Sections

Refer to protective measures listed in Sections 8 and 13.

## SECTION 7: HANDLING AND STORAGE

### Precautions for Safe Handling

Ensure adequate ventilation. Wear personal protective equipment/face protection. Avoid ingestion and inhalation. Avoid contact with skin, eyes or clothing. Avoid dust formation.

### Conditions for Safe Storage, Including any Incompatibilities

Keep container tightly closed in a dry and well-ventilated place. Protect from moisture.

### Specific End Uses

Use in laboratories.

## SECTION 8: EXPOSURE CONTROLS/PERSONAL PROTECTION

### Control Parameters

### Exposure Controls

#### Engineering Measures

Ensure adequate ventilation, especially in confined areas. Ensure that eyewash stations and safety showers are close to the workstation location.

### Personal protective equipment

#### **Eye Protection**

Wear safety glasses with side shields (or goggles)

#### **Hand Protection**

Protective gloves

#### **Skin and body protection**

Wear appropriate protective gloves and clothing to prevent skin exposure

Inspect gloves before use.

Please observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information)

Ensure gloves are suitable for the task: Chemical compatibility, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion.

Remove gloves with care avoiding skin contamination.

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**Respiratory Protection** When workers are facing concentrations above the exposure limit they must use appropriate certified respirators  
**Recommended Filter type:** Particle filter

**Hygiene Measures** Handle in accordance with good industrial hygiene and safety practice

**Environmental exposure controls** No information available

## SECTION 9: PHYSICAL AND CHEMICAL PROPERTIES

### Information on basic physical and chemical properties

|                                     |                          |                                          |
|-------------------------------------|--------------------------|------------------------------------------|
| <b>Appearance</b>                   | White                    |                                          |
| <b>Physical State</b>               | Solid                    |                                          |
| <b>Odor</b>                         | Odorless                 |                                          |
| <b>Odor Threshold</b>               | No data available        |                                          |
| <b>pH</b>                           | 4.4-4.6 @ 20°C           | (10 g/l aq.sol)                          |
| <b>Melting Point/Range</b>          | 60 °C / 140 °F           |                                          |
| <b>Softening Point</b>              | No data available        |                                          |
| <b>Boiling Point/Range</b>          | No information available |                                          |
| <b>Flash Point</b>                  | No information available | <b>Method -</b> No information available |
| <b>Evaporation Rate</b>             | Not applicable           | Solid                                    |
| <b>Flammability (solid,gas)</b>     | No information available |                                          |
| <b>Explosion Limits</b>             | No data available        |                                          |
| <b>Vapor Pressure</b>               | No data available        |                                          |
| <b>Vapor Density</b>                | Not applicable           | Solid                                    |
| <b>Specific Gravity / Density</b>   | No data available        |                                          |
| <b>Bulk Density</b>                 | No data available        |                                          |
| <b>Water Solubility</b>             | Soluble                  |                                          |
| <b>Solubility in other solvents</b> | No information available |                                          |

### Partition Coefficient (n-octanol/water)

|                                  |                          |       |
|----------------------------------|--------------------------|-------|
| <b>Autoignition Temperature</b>  | No data available        |       |
| <b>Decomposition Temperature</b> | No data available        |       |
| <b>Viscosity</b>                 | Not applicable           | Solid |
| <b>Explosive Properties</b>      | No information available |       |
| <b>Oxidizing Properties</b>      | No information available |       |

**Molecular Formula** H2 Na O4 P . 2 H2 O  
**Molecular Weight** 156.01

## SECTION 10: STABILITY AND REACTIVITY

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## Reactivity

None known, based on information available.

## Chemical Stability

Stable under normal conditions.

## Possibility of Hazardous Reactions

### **Hazardous Polymerization Hazardous Reactions**

Hazardous polymerization does not occur.  
None under normal processing.

## Conditions to Avoid

Incompatible products. Temperatures above 169°C. Avoid dust formation. Exposure to moist air or water.

## Incompatible Materials

Strong oxidizing agents. Strong acids. Strong bases.

## Hazardous Decomposition Products

Oxides of phosphorus.

## SECTION 11: TOXICOLOGICAL INFORMATION

### Information on Toxicological Effects

#### **Product Information**

#### **(a) acute toxicity;**

**Oral**

Based on available data, the classification criteria are not met

**Dermal**

Based on available data, the classification criteria are not met

**Inhalation**

Based on available data, the classification criteria are not met

| Component                   | LD50 Oral                 | LD50 Dermal                  | LC50 Inhalation              |
|-----------------------------|---------------------------|------------------------------|------------------------------|
| Sodium phosphate, monobasic | LD50 = 8290 mg/kg ( Rat ) | LD50 > 7940 mg/kg ( Rabbit ) | LC50 > 0.83 mg/L ( Rat ) 4 h |

#### **(b) skin corrosion/irritation;**

Based on available data, the classification criteria are not met

#### **(c) serious eye damage/irritation;**

Based on available data, the classification criteria are not met

#### **(d) respiratory or skin sensitization;**

**Respiratory**

Based on available data, the classification criteria are not met

**Skin**

Based on available data, the classification criteria are not met

#### **(e) germ cell mutagenicity;**

Based on available data, the classification criteria are not met

#### **(f) carcinogenicity;**

Based on available data, the classification criteria are not met

There are no known carcinogenic chemicals in this product

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|                                           |                                                                                                                                                            |
|-------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|
| (g) reproductive toxicity;                | Based on available data, the classification criteria are not met                                                                                           |
| (h) STOT-single exposure;                 | Based on available data, the classification criteria are not met                                                                                           |
| (i) STOT-repeated exposure;               | Based on available data, the classification criteria are not met                                                                                           |
| Target Organs                             | None known.                                                                                                                                                |
| (j) aspiration hazard;                    | Not applicable<br>Solid                                                                                                                                    |
| Other Adverse Effects                     | May cause respiratory irritation May be harmful if swallowed                                                                                               |
| Symptoms / effects,both acute and delayed | May be harmful if absorbed through skin. May cause adverse kidney effects. Ingestion may cause gastrointestinal irritation, nausea, vomiting and diarrhea. |
| Endocrine Disrupting Properties           | Assess endocrine disrupting properties for human health. This product does not contain any known or suspected endocrine disruptors.                        |

## SECTION 12: ECOLOGICAL INFORMATION

|                                        |                                                                                                                                                              |
|----------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <u>Ecotoxicity effects</u>             | Do not empty into drains. Do not flush into surface water or sanitary sewer system.                                                                          |
| <u>Persistence and degradability</u>   | Readily biodegradable                                                                                                                                        |
| Persistence                            | Soluble in water, Persistence is unlikely, based on information available.                                                                                   |
| Degradability                          | Not relevant for inorganic substances.                                                                                                                       |
| <u>Bioaccumulative potential</u>       | Bioaccumulation is unlikely                                                                                                                                  |
| <u>Mobility in soil</u>                | The product is water soluble, and may spread in water systems. Will likely be mobile in the environment due to its water solubility. Highly mobile in soils. |
| <u>Endocrine Disruptor Information</u> | This product does not contain any known or suspected endocrine disruptors                                                                                    |
| <u>Other adverse effects</u>           | No information available                                                                                                                                     |

## SECTION 13: DISPOSAL CONSIDERATIONS

|                                     |                                                                                               |
|-------------------------------------|-----------------------------------------------------------------------------------------------|
| <u>Waste treatment methods</u>      |                                                                                               |
| Waste from Residues/Unused Products | Dispose of in accordance with local regulations                                               |
| Contaminated Packaging              | Empty containers should be taken to an approved waste handling site for recycling or disposal |

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## SECTION 14: TRANSPORT INFORMATION

|                                     |                                 |
|-------------------------------------|---------------------------------|
| <b>IMDG/IMO</b>                     | Not regulated                   |
| <b>Road and Rail Transport</b>      | Not regulated                   |
| <b>IATA</b>                         | Not regulated                   |
| <b>Special Precautions for User</b> | No special precautions required |

## SECTION 15: REGULATORY INFORMATION

### Safety, health and environmental regulations/legislation specific for the substance or mixture

**International Inventories** X = listed

| Component                             | EINECS    | TSCA | DSL | PICCS | ENCS | ISHL | IECSC | AICS | KECL     |
|---------------------------------------|-----------|------|-----|-------|------|------|-------|------|----------|
| Sodium phosphate, monobasic dihydrate | -         | -    | -   | X     | X    |      | X     | X    | -        |
| Sodium phosphate, monobasic           | 231-449-2 | X    | X   | X     | X    | X    | X     | X    | KE-31577 |

### National Regulations

**Persistent Organic Pollutant** This product does not contain any known or suspected substance  
**Ozone Depletion Potential** This product does not contain any known or suspected substance

## SECTION 16: OTHER INFORMATION

### Legend

**CAS** - Chemical Abstracts Service

**EINECS/ELINCS** - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

**PICCS** - Philippines Inventory of Chemicals and Chemical Substances

**IECSC** - Chinese Inventory of Existing Chemical Substances

**KECL** - Korean Existing and Evaluated Chemical Substances

**WEL** - Workplace Exposure Limit

**ACGIH** - American Conference of Governmental Industrial Hygienists

**RPE** - Respiratory Protective Equipment

**LC50** - Lethal Concentration 50%

**POW** - Partition coefficient Octanol:Water

**TSCA** - United States Toxic Substances Control Act Section 8(b) Inventory

**DSL/NDSL** - Canadian Domestic Substances List/Non-Domestic Substances List

**ENCS** - Japanese Existing and New Chemical Substances

**AICS** - Australian Inventory of Chemical Substances

**NZIoC** - New Zealand Inventory of Chemicals

**TWA** - Time Weighted Average

**IARC** - International Agency for Research on Cancer

**LD50** - Lethal Dose 50%

**EC50** - Effective Concentration 50%

**ADR** - European Agreement Concerning the International Carriage of Dangerous Goods by Road

**IMO/IMDG** - International Maritime Organization/International Maritime Dangerous Goods Code

**OECD** - Organisation for Economic Co-operation and Development

**BCF** - Bioconcentration factor

**ICAO/IATA** - International Civil Aviation Organization/International Air Transport Association

**MARPOL** - International Convention for the Prevention of Pollution from Ships

**ATE** - Acute Toxicity Estimate

**VOC** - (Volatile Organic Compound)

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## Key literature references and sources for data

<https://echa.europa.eu/information-on-chemicals>

Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

Revision Date

22-Mar-2025

Revision Summary

Not applicable.

**In accordance with local and national regulations: Occupational Safety and Health  
(Classification, Labelling and Safety Data Sheet of Hazardous Chemicals) Regulations 2013**

## Disclaimer

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of Safety Data Sheet**